Large-Scale Polygonal Building Mapping From 3–5 m Medium-Resolution Satellite Imagery

Fahong Zhang and Xiao Xiang Zhu, *Fellow, IEEE*

***Abstract*—Mapping buildings in vectorized form from remote sensing imagery is a key task for digitalizing and monitoring our planet. While most existing methods rely on sub-meter-resolution aerial or satellite imagery to extract fine-grained building polygons, these sources suffer from long revisit intervals, high cost, and incomplete coverage, limiting their scalability. This raises a fundamental question: can we enable global-scale building polygon mapping using only medium-resolution (3–5 m) satellite imagery, while maintaining acceptable geometric quality? In this article, we explore the feasibility of large-scale polygonal building mapping using medium-resolution (approximately 4.77 m at the equator) PlanetScope imagery. Our goal is not to surpass high-resolution methods in geometric precision, but to achieve substantially improved coverage and update frequency with a favorable accuracy–scalability trade-off. To this end, we present a complete and scalable pipeline encompassing global data acquisition, automated annotation, novel algorithmic components, and an efficient inference framework tailored for real-world deployment. To address the common issue of small, adjacent buildings merging during binarization, we introduce peak-based clustering (PBC) for robust instance separation. In addition, we propose a point feature sampling (PFS) strategy that extends the global collinearity-aware polygonization (GCP) method to efficiently generate regularized building polygons from large-scale satellite mosaics. Extensive experiments show that our approach achieves substantially higher completeness than existing high-resolution-based products, with only minimal accuracy trade-offs. These results highlight the potential of our method to significantly enhance global building footprint coverage. The code and data are publicly released under: https://github.com/zhu-xlab/MRPolyBuild.**

***Index Terms*—Building polygonal mapping, building polygonization, dynamic programming, instance segmentation.**

I. Introduction

ACCURATE and up-to-date polygonal building footprint maps, which delineate the vector outlines of individual structures, are critical for structured spatial analysis in domains, such as urban planning, cadastral mapping, and infrastructure management. In recent years, several continent-scale polygonal building footprint datasets have been released, including Microsoft Building Footprints (MBFs) [1], which covers most of the world except China; Google Open Buildings (GOBs) [2], which focuses on the Global South; and the comprehensive large-scale mapping (CLSM) dataset [3], which targets East Asia. These datasets have greatly expanded global access to vectorized building information.

Nevertheless, they are all derived from high-resolution satellite imagery, which is expensive to acquire and constrained by limited revisit frequency. As a result, their update cycles can span several years, and regions with persistent cloud cover, such as tropical rainforests and coastal zones, often remain underrepresented due to continuous data gaps. Another challenge with high-resolution satellite imagery is the frequent use of off-nadir viewing angles. This leads to parallax effects, especially for tall buildings, where rooftops appear shifted relative to their actual ground locations. Consequently, the extracted building polygons often suffer from systematic misalignment and positional offsets.

In this work, we explore a fundamental question: Is it possible to perform fine-grained, global-scale building mapping and polygonization using lower-resolution satellite imagery with higher revisit frequency, enabling more frequent updates? PlanetScope imagery [4] offers near-daily global coverage at 3–5 m resolution, and its Basemap product provides global mosaics at quarterly or shorter intervals. In this study, we use the 3-band 8-bit RGB Basemap product. While promising, this line of research presents several technical challenges that must be addressed.

First, constructing a global-scale training dataset is challenging. Most publicly available building datasets are derived from high-resolution imagery and are typically restricted to city-scale regions. Such datasets are inadequate for training models that generalize effectively to medium-resolution imagery at continental or global scales. Second, buildings in medium-resolution images often occupy only a few pixels, which causes semantic segmentation models to merge nearby buildings into a single connected component, making instance-level delineation difficult. Third, conventional deep learning pipelines are generally designed for small image patches, such as 1024×1024 pixels, whereas the satellite scenes used in this study can reach sizes up to $32\,768 \times 32\,768$ pixels. This substantial scale difference calls for new strategies to enable efficient and scalable polygon extraction.

To address these challenges, we propose a unified framework with the following components. First, we construct a large-scale training dataset by fusing multiple existing polyg-

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Fahong Zhang is with the Chair of Data Science in Earth Observation, Technical University of Munich, 80333 Munich, Germany (e-mail: fahong.zhang@tum.de).

Xiao Xiang Zhu is with the Chair of Data Science in Earth Observation, Technical University of Munich, 80333 Munich, Germany, and also with Munich Center for Machine Learning, 80333 Munich, Germany (e-mail: xiaoxiang.zhu@tum.de).

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onal building footprint sources, including OpenStreetMap (OSM) [5], MBF, GOB, and CLSM, and sampling training patches across the globe using a clustering-based strategy. This yields a globally diverse annotation set covering approximately 200 000 km^2 of land. Second, we propose a robust instance separation method called peak-based clustering (PBC), which improves over standard binarization techniques by more effectively separating small adjacent building instances in noisy segmentation maps. Third, we introduce a novel point feature sampling (PFS) strategy that enables the previously proposed global collinearity-aware polygonization (GCP) [6] method to process large-scale scenes efficiently. Finally, by utilizing PlanetScope's quarterly Basemap data, our framework can generate global building footprint maps at a three-month update frequency if computation resources are satisfied. Our contributions can be summarized as follows.

1) To the best of our knowledge, this is the first work to demonstrate the feasibility of detecting and vectorizing buildings at a large scale using satellite imagery with a spatial resolution below 3 m.
2) We construct a medium-resolution, global-scale building polygon dataset by fusing existing sources and sampling across diverse geographies.
3) We develop PBC, a robust instance separation technique that improves detection of small, closely packed buildings.
4) We demonstrate that our framework achieves higher spatial coverage than existing continent-scale high-resolution products, with an affordable accuracy trade-off.

II. Related Works

A. Polygonal Building Mapping

Polygonal building mapping aims to localize buildings in remote sensing imagery and generate their corresponding vectorized polygon representations. Existing methods can be broadly categorized into segmentation-based and instance modeling-based approaches depending on their prediction pipelines.

1) Segmentation-Based Methods: These methods first apply semantic segmentation models to predict per-pixel building masks or auxiliary geometric representations, such as vertex masks. To extract polygonal outlines, one common decoding strategy is to apply contour tracing algorithms such as marching squares [7] on the predicted binary masks. Methods like frame field learning [8] follow this approach, and often apply refinement techniques to simplify and regularize the resulting polygons.

Other methods, such as PolyWorld [9], HiSup [10], and Pix2Poly [11], take a vertex-based representation, where a network predicts building vertices and graph connectivity to assemble the final polygonal contours [12].

2) Instance Modeling-Based Methods: These methods explicitly model individual buildings through object detection or instance segmentation frameworks, such as Mask R-CNN [13], DETR [14], and Mask2Former [15]. Polygon decoding is then applied on each predicted instance via different strategies.

The first strategy involves generating a binary mask for each instance and converting it into a polygon. For example, [16] uses Mask R-CNN to predict instance masks, followed by minimum description length (MDL) optimization [17] to regularize the contours. GCP [6] also follows this paradigm, applying a polyline refinement network and a collinearity-aware simplification module to generate compact and accurate vector polygons.

The second strategy formulates polygon prediction as a sequence generation problem, where models like Polygon-RNN [18], PolyMapper [19], HiT [20], and P2PFormer [21] predict building corners sequentially using RNNs or Transformers.

Other methods, such as PolyBuilding [22], predict a fixed set of contour points per building and apply a filtering module to prune redundant vertices and generate the final polygon.

3) Discussion: The above methods, while effective on high-resolution imagery [23], face notable challenges when applied to large-scale polygonal building mapping using medium-resolution satellite data. For segmentation-based methods, detecting building corners or vertices becomes unreliable due to the coarse spatial resolution. Mask-to-polygon conversion is a more feasible alternative, but care must be taken to avoid merging adjacent buildings during the binarization process, which is a common issue in dense urban scenes. For instance, modeling-based methods, scalability is a major concern. A single 256×256 patch from 3-m-resolution satellite imagery may contain thousands of small buildings, making instance-level detection computationally expensive.

These limitations highlight the need for methods that are both scalable and robust to medium-resolution imagery, while preserving the quality of polygon-level outputs.

B. Large-Scale Vectorized Building Polygon Datasets

Although the above-introduced polygonal building mapping methods have achieved impressive results on public benchmarks, few of them have been deployed to generate building footprints at national or continental scales. As a result, large-scale, high-quality vectorized building datasets remain limited.

Several publicly available large-scale polygon products exist. MBFs [1] provides building polygons for most of the world, excluding China. The dataset is derived from high-resolution satellite imagery using a deep learning-based segmentation and post-processing pipeline. It is updated infrequently and mainly reflects data collected between 2014 and 2022. GOBs [2] focuses on the Global South region, including Africa, parts of Asia, and Latin America. It is based on submeter resolution imagery and uses a semantic segmentation framework. Similar to Microsoft, the dataset does not cover China or East Asia. To address this gap, CLSM [3] offers building footprint data for five East Asian countries, generated from Google Earth images. Other global-scale polygonal products, such as GlobalBuildingAtlas [24] and 3-D-GloBFP [25], have been created by fusing other sources. However, these merged datasets still inherit the limitations of their original sources.

Despite their global scope, current large-scale vectorized building datasets face several limitations.

1) *Low Temporal Resolution:* All existing products rely on high-resolution imagery, which is costly and updated infrequently due to low revisit frequency.
2) *Coverage Gaps:* Cloud-prone areas such as tropical rainforests often lack usable imagery, resulting in data voids.
3) *Geometric Inaccuracy:* High-rise buildings in off-nadir imagery may exhibit positional errors or distorted outlines due to occlusion and perspective.
4) *High Computational and Storage Costs:* Global-scale processing of high-resolution imagery demands substantial computational and storage resources.

These limitations motivate alternative approaches based on more frequently updated medium-resolution satellite imagery, as discussed in Section II-C.

C. Large-Scale Raster-Based Building Mapping

Another related line of work focuses on large-scale building or built-up mapping from freely available satellite imagery, but mainly produces raster products rather than instance-level vector polygons. For example, [26] introduces the Open Buildings 2.5-D Temporal Dataset, which is generated from multitemporal 10 m Sentinel-2 imagery using a teacher–student super-resolution framework, providing annual raster products of building presence, fractional building count, and building height [27]. Beyond building-level mapping, large-scale settlement and built-up products, such as World Settlement Footprint [28], GHSL/GHS-BUILT [29], and Global Urban Footprint [30] provide global raster maps of human settlements or built-up areas.

These works demonstrate the potential of freely available satellite imagery for scalable and temporally frequent building and built-up monitoring. However, their outputs are typically limited to raster-level information, such as building presence, built-up extent, density, or height, rather than instance-level vector polygons with explicit building boundaries. In contrast, our work targets refined vectorized building footprint generation from medium-resolution imagery, aiming to preserve the scalability and temporal advantages of such imagery while producing GIS-ready polygon outputs.

III. Methods

This section presents the overall pipeline for generating building polygons using medium-resolution satellite images.

A. Overview

The overall inference pipeline is shown in Fig. 1. A large satellite image is first divided into overlapping patches, each processed by a semantic segmentation model to produce building probability maps. These maps are merged into a unified probability map via Gaussian-weighted averaging, which emphasizes patch centers and suppresses edge artifacts.

Building instances are extracted using a PBC algorithm that identifies local maxima and groups nearby pixels. Each instance is converted into a binary mask and vectorized using OpenCV's contour tracing [31].

To enhance contour quality, a polyline refinement module adjusts vertex positions using features from the PFS module, which captures spatial and texture cues. The refined polylines are then simplified by the GCP module [6], producing compact and regularized building polygons.

B. PBC Module

The goal of the proposed PBC module is to convert the building probability map or confidence map $\mathbf{C} \in \mathbb{R}^{H\times W}$, predicted by a semantic segmentation model, into a discrete instance mask $\mathbf{M} \in \mathbb{N}^{H\times W}$, where each nonzero value indicates the index of a detected building instance.

Traditional approaches, including the method used in [2], typically binarize $\mathbf{C}$ using a fixed threshold and extract connected components to obtain $\mathbf{M}$. However, in densely built areas, the probability values between adjacent buildings are often high, making threshold-based methods prone to merging multiple buildings into a single instance.

To address this issue, the PBC module is designed based on the observation that pixels near the center of a building generally exhibit higher probability values than those near the boundary. Consequently, each building instance typically forms a local "peak" in the probability map. By identifying and growing from these peaks, we can more effectively separate adjacent buildings.

The PBC algorithm proceeds by first sorting all pixels in $\mathbf{C}$ in descending order of their probability values. It then iterates over each pixel in this order, assigning instance labels based on the state of its neighboring pixels. Specifically, there will be three different cases.

1) If the pixel has no previously labeled neighbors, it is considered a local peak. A new cluster is created, and the pixel is assigned a new instance ID.
2) If the pixel has exactly one labeled neighbor, it is assigned to the same cluster as that neighbor.
3) If a pixel is adjacent to multiple clusters, we compute the probability gap between it and the lowest neighboring peak. If the gap exceeds a threshold ξ, the pixel is assigned to one of the clusters, and all adjacent clusters are marked inactive to prevent future pixels from being assigned to them, thus avoiding spatially touching between clusters. Otherwise, the clusters are merged, and the pixel is added to the merged cluster.

In terms of efficiency, the PBC algorithm has a time complexity of $O(n \log n)$, where $n = H \times W$ is the number of pixels. This is dominated by the initial sorting step. The subsequent per-pixel iteration and cluster merging are implemented using a Union-Find data structure with path compression, enabling near-linear $O(n)$ performance for label assignment. Algorithm 1 provides the pseudocode of PBC.

The proposed PBC is related to classical peak-based clustering, region growing, and watershed segmentation [32], but is designed as an instance decoding module for building probability maps rather than a generic clustering method. Unlike standard region growing or watershed algorithms, PBC does not require manually specified seeds or a predefined number of instances. Instead, it processes pixels in descending

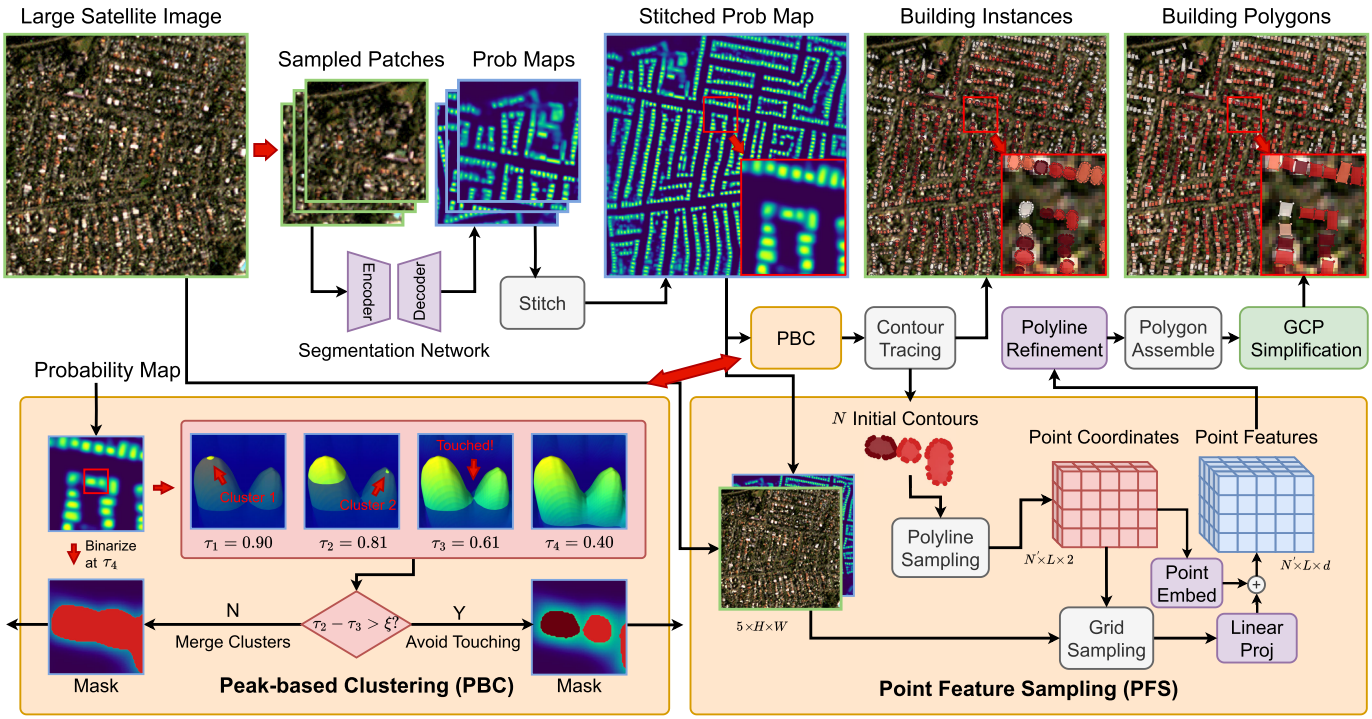


Fig. 1. Illustration of the proposed polygonal building mapping framework. Purple boxes indicate neural network modules with learnable parameters. Green boxes represent nonparametric operations executed on the GPU, while gray boxes correspond to CPU-based processing. Satellite image © Planet Labs.

confidence order and automatically uses local maxima as seeds.

Another key difference is the probability-gap criterion used when two growing regions meet. If the gap between their peak confidences is small, the regions are merged; otherwise, they are kept separate, and the involved clusters are marked inactive to avoid later low-confidence pixels bridging adjacent buildings. This is particularly useful in medium-resolution imagery, where neighboring buildings often have weak separations and are easily merged by simple thresholding or connected-component extraction. Therefore, PBC serves as an efficient decoding step that converts CNN-predicted probability maps into separated building instances before polygon vectorization.

C. Point Feature Sampling

Given the instance mask $\mathbf{M}$ produced by the PBC module, we first vectorize each connected region into N initial contours and uniformly resample them into fixed-length polylines $\mathbf{P} \in \mathbb{R}^{N' \times L \times 2}$, following the method proposed in [6]. These polylines offer structured approximations of building boundaries but are often coarse or misaligned due to the limited spatial resolution of the input.

To enable vertex refinement via neural networks, we extract local features for each polyline point using a differentiable sampling mechanism. Specifically, we construct a five-channel feature map $\mathbf{F} \in \mathbb{R}^{H \times W \times 5}$ by concatenating the RGB image $\mathbf{x}$, the building probability map $\mathbf{C}$, and its complement $1 - \mathbf{C}$. For each vertex $\mathbf{P}_{i,j} \in \mathbb{R}^2$, we define a local $k \times k$ floating-point grid centered at the vertex and apply bilinear interpolation over $\mathbf{F}$ using a differentiable grid sampling operator (e.g., `grid_sample` in PyTorch). This yields a local patch feature, resulting in a structured sequence $\mathbf{Z} \in \mathbb{R}^{N' \times L \times (5k^2)}$.

Meanwhile, we encode each vertex by normalizing its 2-D coordinates and passing them through a linear layer to obtain a point embedding. The final feature representation $\mathbf{Z}' \in \mathbb{R}^{N' \times L \times d}$ is obtained by summing these point embeddings with the projected patch features after a linear transformation.

D. Polyline Refinement

To refine the polylines $\mathbf{P}$, we feed the extracted point features $\mathbf{Z}'$ into a Transformer decoder [15] that predicts vertex-wise offsets. The decoder follows a three-layer architecture, where each layer comprises a self-attention block, a cross-attention block, and a feed-forward network. The refined polyline $\tilde{\mathbf{P}}$ is obtained by applying the predicted offsets to the initial polyline $\mathbf{P}$

$$\tilde{\mathbf{P}} = \mathbf{P} + \text{TransformerDecoder}\left(\mathbf{Z}'\right). \tag{1}$$

E. GCP Simplification

Following the method proposed in GCP [6], the refined polylines $\tilde{\mathbf{P}}$ are assembled into building polygons and subsequently simplified by removing redundant vertices. This process produces the final compact and clean polygon outputs.

F. Loss Functions

To compute the training loss, we consider all predicted and ground-truth polygon pairs whose IoU exceeds 0.3 as matched pairs. Each matched ground-truth polygon is converted into a fixed-length polyline using the same uniform resampling procedure introduced in PFS.

Algorithm 1 PBC Algorithm

Input: $\mathbf{C} \in \mathbb{R}^{H\times W}$: probability map; threshold ξ
Output: $\mathbf{M} \in \mathbb{N}^{H\times W}$: instance mask
Initialize $\mathbf{M} \leftarrow 0^{H\times W}$, $\mathbf{V} \leftarrow \texttt{false}^{H\times W}$ // instance labels and valid-pixel mask;
Initialize `parent`, `peak_value`, `touched` as empty maps // cluster parent, reference peak, and valley flag;
`next_id` $\leftarrow 1$;
Sort all pixels (i, j) by $\mathbf{C}_{i,j}$ descending into list $\mathcal{P}$ // high-confidence pixels are processed first;
Function `Find`(x)**:**
  if `parent`$[x] \neq x$ **then**
    `parent`$[x] \leftarrow$ `Find(parent`$[x]$`)` // path compression in Union-Find algorithm;
  return `parent`$[x]$;
Function `Merge`$(l_0, \mathcal{L})$**:**
  foreach $l \in \mathcal{L}$ **do**
    $r_l \leftarrow$ `Find`(l);
    $r_0 \leftarrow$ `Find`(l_0);
    if $r_l \neq r_0$ **then**
      `parent`$[r_l] \leftarrow r_0$ // merge two cluster roots;
      `peak_value`$[r_0] \leftarrow \min\{$`peak_value`$[r_0],$ `peak_value`$[r_l]\}$ // retain the lower reference peak;
      `touched`$[r_0] \leftarrow$ `touched`$[r_0] \vee$ `touched`$[r_l]$ // propagate the valley flag;
foreach $(i, j, p) \in \mathcal{P}$ **do**
  $\mathcal{N} \leftarrow$ 8-connected neighbors of (i, j);
  $\mathcal{L} \leftarrow \{\text{Find}(\mathbf{M}_{u,v}) \mid (u, v) \in \mathcal{N},\ \mathbf{M}_{u,v} > 0\}$ // labels of neighboring clusters;
  if $|\mathcal{L}| = 0$ **then**
    $\mathbf{M}_{i,j} \leftarrow$ `next_id` // create a cluster at a new peak;
    `parent`[`next_id`] $\leftarrow$ `next_id`;
    `peak_value`[`next_id`] $\leftarrow p$;
    `touched`[`next_id`] $\leftarrow$ `false`;
    `next_id` $\leftarrow$ `next_id` $+ 1$;
  else
    if $|\mathcal{L}| = 1$ **then**
      $l \leftarrow$ the only label in $\mathcal{L}$;
      $\mathbf{M}_{i,j} \leftarrow l$ // expand the neighboring cluster;
      if `touched`$[l] =$ `true` **then**
        $\mathbf{V}_{i,j} \leftarrow$ `false` // reject later low-confidence pixels;
    else
      $p_{\min} \leftarrow \min\{$`peak_value`$[l] \mid l \in \mathcal{L}\}$;
      $\delta \leftarrow p_{\min} - p$ // peak-to-current probability gap;
      Choose $l_0 \in \mathcal{L}$ as root label;
      $\mathbf{M}_{i,j} \leftarrow l_0$;
      if $\delta > \xi$ **then**
        foreach $l \in \mathcal{L}$ **do**
          `touched`$[l] \leftarrow$ `true` // prevent later bridge connections;
      else
        `Merge`$(l_0, \mathcal{L})$ // merge across a shallow valley;
        if `touched`$[l_0] =$ `true` **then**
          $\mathbf{V}_{i,j} \leftarrow$ `false`;
foreach $(i, j, p) \in \mathcal{P}$ **do**
  if $\mathbf{V}_{i,j} =$ `false` **then**
    $\mathbf{M}_{i,j} \leftarrow 0$ // remove invalid bridge pixels;

Let $\tilde{\mathbf{p}} \in \mathbb{R}^{L\times 2}$ denote a predicted polyline and $\mathbf{p}_{\text{gt}} \in \mathbb{R}^{L'\times 2}$ its corresponding ground-truth polyline. Let $\tilde{\mathbf{p}}^*$ be the simplified version of $\tilde{\mathbf{p}}$ obtained using the GCP simplification algorithm. The overall loss is formulated as follows:

$$\mathcal{L} = \alpha\mathcal{L}_{\text{vert}}\left(\tilde{\mathbf{p}}, \mathbf{p}_{\text{gt}}\right) + \beta\mathcal{L}_{\text{gc}}\left(\tilde{\mathbf{p}}, \tilde{\mathbf{p}}^*\right) + \gamma\mathcal{L}_{\text{rht}}\left(\tilde{\mathbf{p}}, \mathbf{p}_{\text{gt}}\right) \quad (2)$$

where both $\mathcal{L}_{\text{vert}}$ and $\mathcal{L}_{\text{gc}}$ are adopted from [6], with the former supervising vertex-level alignment between predicted and ground truth polygons, and the latter promoting piecewise linearity through a global collinearity prior. $\mathcal{L}_{\text{rht}}$ additionally penalizes deviations from right angles when applicable. The weights α, β, and γ control the relative contribution of each loss term.

To promote geometric regularity and align predicted polygons with dominant structural directions commonly found in buildings, we introduce a right-angle loss $\mathcal{L}_{\text{rht}}$ that penalizes deviations from orthogonal alignment. The computation involves the following steps.

1) *Canonical Base Discretization:* We define $K = 16$ pairs of orthogonal vectors $\{(\mathbf{x}_k, \mathbf{y}_k)\}_{k=1}^{K}$, generated by uniformly rotating the canonical basis $(1, 0)$ and $(0, 1)$ over the range $[0°, 90°]$. Each pair represents a candidate dominant orientation and its perpendicular direction.
2) *Dominant Direction Selection:* For each matched ground truth polyline $\mathbf{p}_{\text{gt}}$, we compute the orientation θ_i^{gt} of each edge and evaluate alignment with all K candidate bases. The pair (θ_x, θ_y) that maximizes average cosine similarity is selected as the dominant orthogonal basis.
3) *Loss Computation:* For each edge in the matched predicted polyline $\tilde{\mathbf{p}}$, with orientation θ_i, the right-angle loss

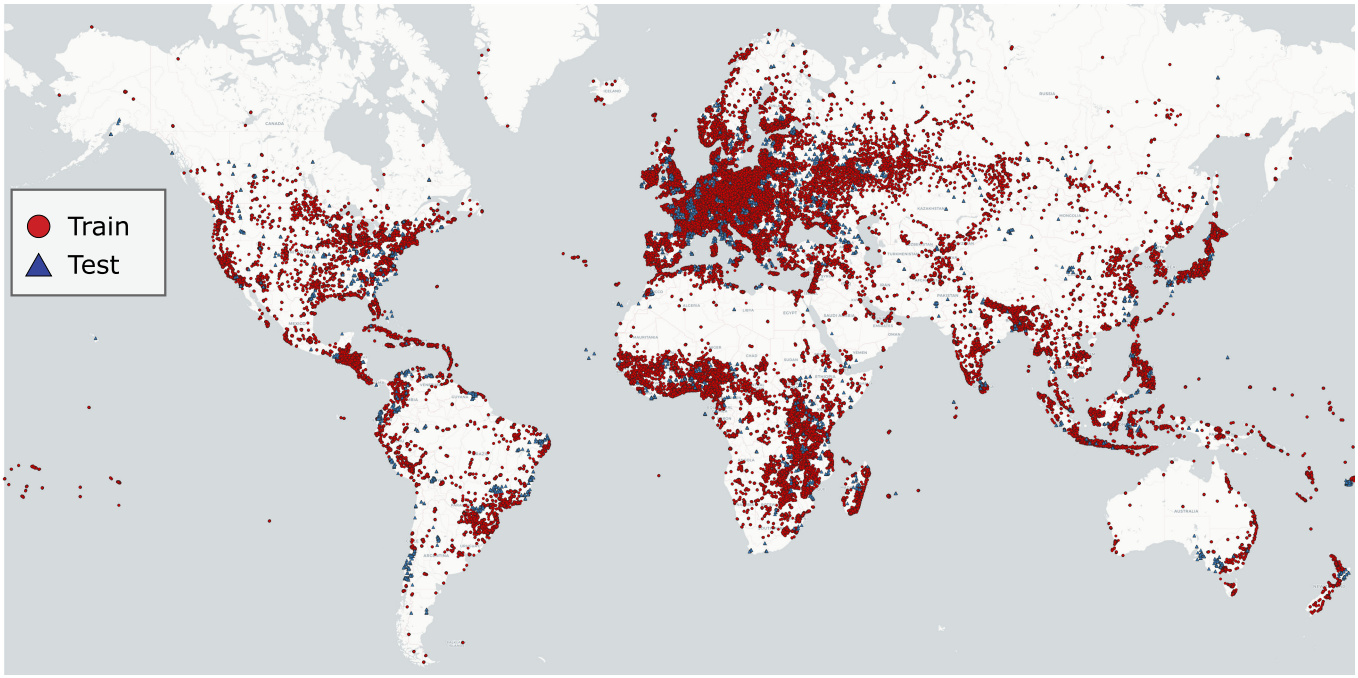


Fig. 2. Spatial distribution of the generated dataset. Basemap @ CartoDB.

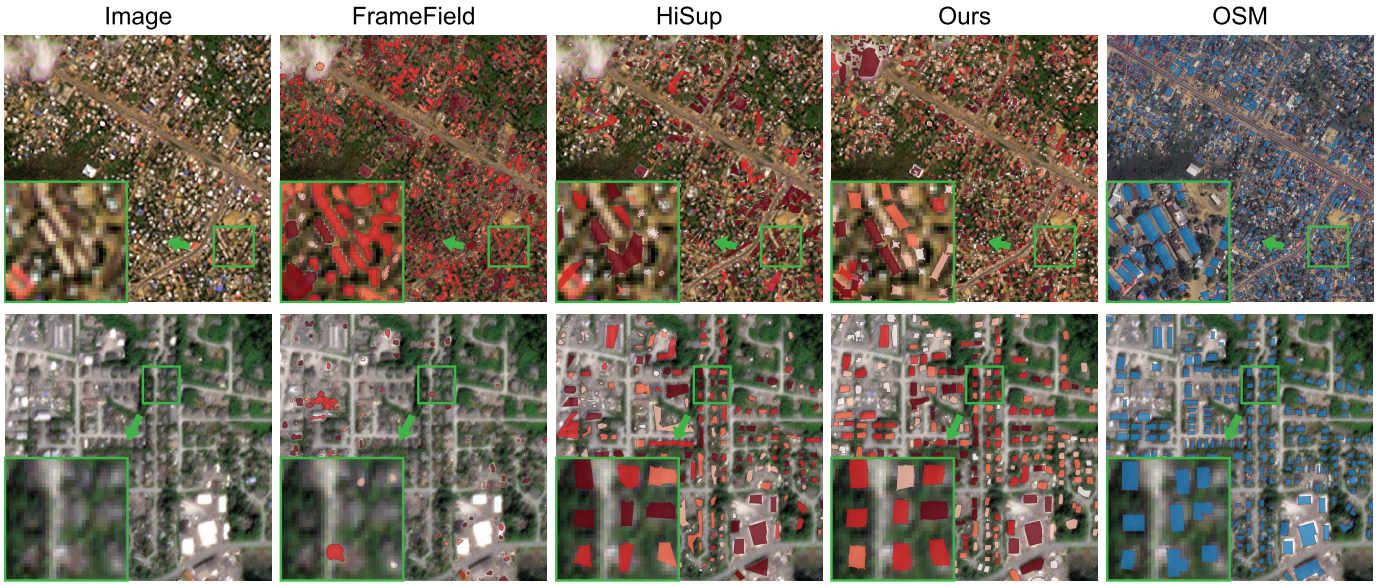


Fig. 3. Visual comparison of polygon extraction results by HiSup, FrameField, and our method on medium-resolution satellite imagery. All methods are trained on the same randomly sampled 10% training subset for 50 epochs. Satellite imagery © Planet Labs PBC and Google Satellite; building annotations © OSM contributors.

measures how well it aligns with either θ_x or θ_y, and is defined as follows:

$$\mathcal{L}_{\text{rht}} = 1 - \frac{1}{L-1}\sum_{i=1}^{L-1} \max\left(\left|\cos(\theta_i - \theta_x)\right|, \left|\cos(\theta_i - \theta_y)\right|\right).$$

This encourages alignment of predicted edges with the dominant horizontal–vertical layout typical in urban building structures.

IV. Experiments

A. Datasets

To train and evaluate our polygon-based building mapping framework at a global scale, we construct a large-scale satellite image dataset with diverse geographic and semantic coverage. The imagery is sourced from the PlanetScope Basemap (Q2 2023), comprising approximately 840 000 globally distributed mosaicked tiles, each of size 4096 × 4096 pixels with a spatial resolution of 4.77 m at the equator. To ensure a balanced distribution across geographic regions and build-

TABLE I

EVALUATION RESULTS OF LARGE-SCALE BUILDING POLYGON PRODUCTS ACROSS REGIONS. "AF," "AS," "EU," "NA," "OC," AND "SA" DENOTE AFRICA, ASIA, EUROPE, NORTH AMERICA, OCEANIA, AND SOUTH AMERICA; "CN" DENOTES CHINA. "RES." INDICATES RESOLUTION. "PREC.," "REC.," AND "COMP." DENOTE PRECISION, RECALL, AND COMPLETENESS

Region	Patch #	Methods	Res.	AP^{poly}_{50}	AP^{poly}_{30}	AR^{poly}_{50}	AR^{poly}_{30}	N-ratio	MTA	PoliS	IoU	Prec.	Rec.	Comp.
AF	807	GOB	0.5m	**12.5**	**29.8**	**25.7**	**45.8**	**1.23**	27.9	2.0	**38.0**	53.4	**56.4**	91.9
		MBF	0.3m	7.7	16.7	19.8	29.5	0.55	**25.0**	**1.6**	34.1	**61.8**	43.3	75.5
		Ours	4.8m	4.0	14.7	10.4	22.9	0.52	33.0	2.7	34.3	50.9	51.2	**100.0**
EU	1,943	GOB	0.5m	-	-	-	-	-	-	-	-	-	-	-
		MBF	0.3m	**8.0**	18.8	**21.0**	**33.4**	**0.65**	**26.6**	**2.8**	**44.5**	**64.4**	**59.0**	84.7
		Ours	4.8m	4.3	**20.7**	13.7	31.5	0.56	31.0	4.4	42.8	63.9	56.6	**100.0**
AS	1,185	GOB	0.5m	4.9	**20.7**	**17.5**	**40.3**	**1.18**	**23.3**	2.5	24.1	50.0	31.6	48.1
		MBF	0.3m	5.4	15.9	15.4	27.9	0.58	25.2	2.0	29.4	54.5	39.0	68.0
		Ours	4.8m	2.3	14.3	10.6	27.5	0.63	29.0	4.0	**37.7**	**54.6**	**54.8**	**100.0**
SA	919	GOB	0.5m	**6.5**	**24.8**	**22.7**	**51.6**	1.82	24.7	3.1	**40.6**	49.1	**68.6**	**100.0**
		MBF	0.3m	3.6	12.1	13.6	27.0	**0.67**	**24.3**	**2.3**	35.5	**51.5**	52.8	88.7
		Ours	4.8m	0.7	8.1	6.0	20.7	0.63	33.0	3.7	32.3	47.3	49.7	**100.0**
NA	842	GOB	0.5m	4.3	14.6	10.6	22.9	0.51	28.4	2.8	18.7	62.3	21.0	15.4
		MBF	0.3m	**23.6**	**31.6**	**32.6**	**38.4**	0.49	**27.1**	**1.3**	**59.4**	**77.3**	**71.8**	93.0
		Ours	4.8m	14.4	30.0	24.5	38.1	**0.58**	32.2	2.6	49.2	66.5	65.3	**100.0**
OC	392	GOB	0.5m	-	-	-	-	-	-	-	-	-	-	-
		MBF	0.3m	**29.2**	**38.7**	**38.6**	44.7	0.53	**28.6**	**1.5**	**43.7**	**77.0**	50.1	54.1
		Ours	4.8m	8.4	36.3	23.9	**51.8**	**0.87**	33.6	3.2	40.9	58.1	**57.8**	**100.0**
CN	307	CLSM	0.5m	2.7	10.2	17.9	36.2	1.63	**19.9**	9.3	27.2	42.7	42.6	99.7
		Ours	4.8m	**8.8**	**27.0**	**26.8**	**48.9**	**1.10**	25.3	**7.2**	**39.6**	**57.9**	**55.4**	**100.0**

ing densities, we assign each tile two metadata attributes: 1) geographic coordinates and 2) the number of buildings annotated in OSM. We then apply k-means clustering over this 2-D feature space to partition the dataset into 2048 clusters. Among them, 500 clusters are randomly selected to form the test set, and the remaining 1548 are used for training. From each cluster, we extract multiple 256 $\times$ 256 image patches that span a range of building densities. This process yields a final dataset containing 162 531 training patches and 6123 test patches, achieving a diverse geographical distribution as illustrated in Fig. 2.

For the evaluation set, we strictly use building polygons from OSM [5] as ground truth. To ensure label quality, test patches are manually filtered to retain only those where at least 80% of buildings are covered by OSM annotations. In contrast, manual filtering is impractical for the training set due to its much larger scale. To enhance the coverage and reliability of training annotations, we incorporate additional large-scale building footprint datasets alongside OSM, including GOBs [2], MBFs [1], and CLSM [3]. Based on empirical evaluation results (refer to Table I in the main paper), we assign a fixed priority to these four sources in the following order: OSM, GOB, MBF, and CLSM, reflecting their relative quality and completeness. For each training patch, all available polygons from the four sources are aggregated, and a filtering step is applied to remove redundant overlaps: when multiple polygons intersect, those from lower-priority sources are discarded. This multisource merging process significantly improves coverage while maintaining consistency across the training set without requiring additional manual post-processing.

Since the training annotations are merged from multiple sources with different acquisition times and annotation quality, the resulting labels may contain noise and spatial inconsistency. We do not explicitly correct these issues in the current version, and leave alignment-aware label refinement for future work.

B. Experimental Settings

1) Implementation Details: We use the ConvNeXt-V2-Base architecture [33] as the backbone for semantic segmentation. Input images are upsampled by a factor of 8 using bilinear interpolation, which significantly improves building extraction accuracy, consistent with observations in [26]. The segmentation network is trained for 50 epochs using the AdamW optimizer [34], with a batch size of 4, an initial learning rate of 10^{-4} (decayed to 10^{-5} after 40 epochs), and a weight decay of 0.05.

The polyline refinement module is trained separately for 320 000 iterations with AdamW. The learning rate is set to 10^{-4} and decayed to 10^{-5} after 256 000 iterations. The loss weights in (2) are set to $\alpha = 1$, $\beta = 0.01$, and $\gamma = 10$. They are used mainly to bring the three loss terms to a similar numerical scale during training. These values were not obtained through extensive manual tuning or cherry-picking. In practice, we found that the refinement performance is not very sensitive to moderate changes in these weights.

For inference, we implement an asynchronous pipeline that parallelizes GPU and CPU operations using multithreading for CPU-bound tasks. All experiments are conducted on 2 NVIDIA L40S GPUs and 2 AMD EPYC 9334 32-core CPUs.

2) Evaluation Metrics: We evaluate polygonal building mapping from multiple perspectives, including detection accu-

TABLE II

AGGREGATED COMPARISON WITH LEARNING-BASED POLYGON EXTRACTION METHODS ON THE GLOBAL UNIFIED VERSION OF THE TEST SET IN SECTION IV-A

Methods	AP^{poly}_{50}	AP^{poly}_{30}	AR^{poly}_{50}	AR^{poly}_{30}	N-ratio	MTA	PoliS	IoU	Prec.	Rec.
FrameField [8]	0.3	2.4	2.3	8.9	**0.87**	55.6	6.9	17.4	26.5	33.4
HiSup [10]	0.7	3.2	2.2	6.1	0.23	43.4	5.0	33.0	49.6	49.6
Ours	**2.0**	**11.8**	**8.2**	**22.1**	0.52	**34.9**	**4.8**	**35.2**	**53.4**	**50.5**

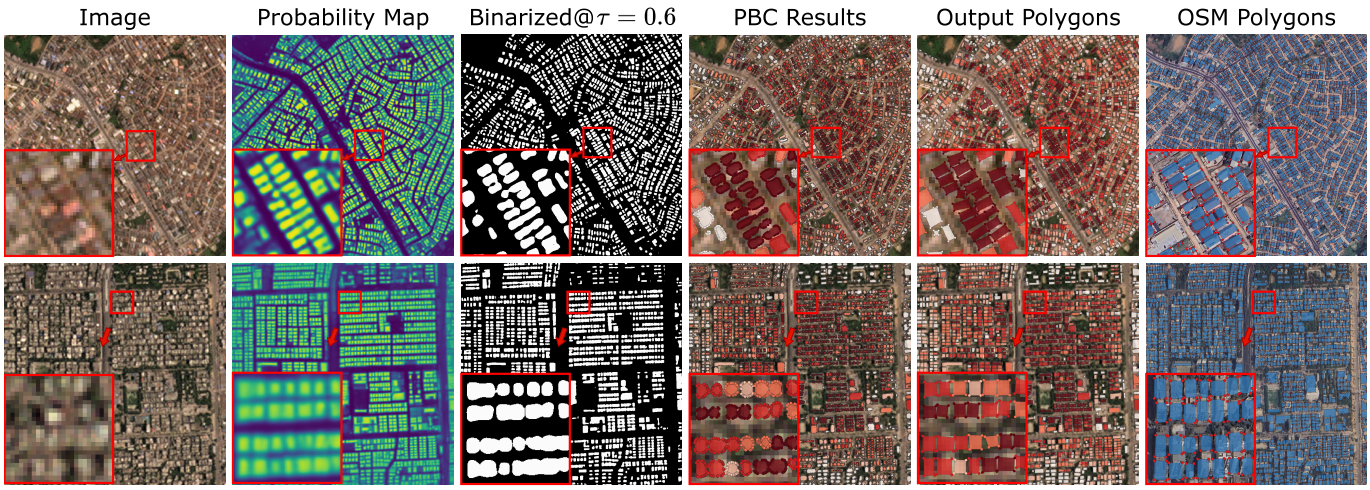


Fig. 4. Visualization of the proposed method under different ablation settings. Satellite imagery © Planet Labs PBC and Google Satellite; building annotations © OSM contributors.

racy, geometric quality, count consistency, and geographic coverage. The following metrics are used.

1) *COCO-Style Metrics:* We report average precision (AP) and average recall (AR) at IoU thresholds of 0.5 and 0.3, with the latter included to account for geo-locational misalignment between remote sensing data and building footprints. Unlike standard COCO evaluation, IoU is computed directly between polygon geometries rather than rasterized masks.
2) *N-Ratio:* Defined as the ratio of predicted to reference polygons, this metric reflects over- or under-detection at a dataset level.
3) *Max Tangent Angle (MTA) error and PoliS:* These metrics evaluate geometric accuracy. MTA [8] measures angular deviation between matched polygon edges (IoU $>$ 0.5), while PoliS [35] computes the average vertex-wise distance between matched polygons.
4) *IoU, Precision, and Recall:* We also report overall geometry-wise IoU, as well as precision and recall scores.
5) *Completeness:* Defined as the percentage of sampled patches whose coordinates fall within the spatial coverage of the evaluated footprint product.

3) Comparison Methods: We compare our method against both existing large-scale building footprint products and learning-based polygon extraction algorithms. For product-level evaluation, we use GOB [2], MBFs [1], and CLSM [3], whose released footprints are directly evaluated without retraining. Since CLSM incorporates OSM annotations in several East Asian countries outside China, we restrict the CLSM comparison to China only.

For algorithm-level evaluation, we compare with frame field learning (FrameField) [8] and HiSup [10], two representative methods originally developed for very-high-resolution satellite imagery. To ensure a fair comparison, both baselines follow the same input preprocessing as our method: images are upsampled by a factor of eight during both training and testing, as described in Section III-A. All other hyperparameters are kept consistent with the original implementations. Because training these polygon extractors on the full global dataset is computationally expensive, FrameField, HiSup, and our method are all trained on the same randomly sampled 10% subset of the training patches in Section III-A for 50 epochs, while all methods are evaluated on the same test set with OSM polygons as reference.

C. Comparative Results

1) Comparison With Existing Footprint Products: We present the evaluation results of different polygonal building footprint products in Table I. In Africa and South America, where GOB offers extensive coverage, it achieves the highest AP^{poly}, AR^{poly}, and IoU scores, showing strong completeness. In Asia, GOB covers parts of South and Southeast Asia and maintains strong COCO-style metrics and N-ratio performance.

In North America, Europe, and Oceania, where GOB coverage is limited, MBF outperforms others in AP and AR at the 0.5 IoU threshold. At a lower threshold of 0.3, our method achieves comparable performance and even surpasses MBF in

TABLE III

ABLATION STUDY RESULTS FOR THE PROPOSED METHOD. τ DENOTES THE THRESHOLD USED FOR BINARIZING THE PROBABILITY MAP. "CT" REFERS TO CONTOUR TRACING, WHILE "PBC" INDICATES THE USE OF PBC TO SEPARATE ADJACENT BUILDING INSTANCES. THE LAST TWO COLUMNS SHOW THE FINAL POLYGON RESULTS WITHOUT AND WITH THE RIGHT-ANGLE LOSS L_{RHT} APPLIED

τ	CT	PBC	GCP w/o $\mathcal{L}_{\text{rht}}$	GCP	AP_{50}^{poly}	AP_{30}^{poly}	AR_{50}^{poly}	AR_{30}^{poly}	N-ratio	MTA	PoliS	IoU	Prec.	Rec.
0.6	✓				2.9	12.6	7.4	19.8	0.48	45.7	4.2	31.1	**74.8**	34.8
0.5	✓				3.7	14.5	10.8	24.7	0.55	45.9	3.9	35.8	68.1	43.0
0.4	✓				**5.2**	12.4	8.7	16.3	0.34	45.8	**3.0**	**44.0**	58.3	**65.2**
0.4	✓	✓			3.6	14.7	12.3	26.9	**0.60**	46.0	3.8	37.7	62.3	48.8
0.4	✓	✓	✓		4.6	**18.3**	**13.7**	29.8	**0.60**	33.9	3.5	40.4	58.8	55.9
0.4	✓	✓		✓	4.5	**18.3**	**13.7**	**29.9**	**0.60**	**31.5**	3.5	40.5	58.7	56.5

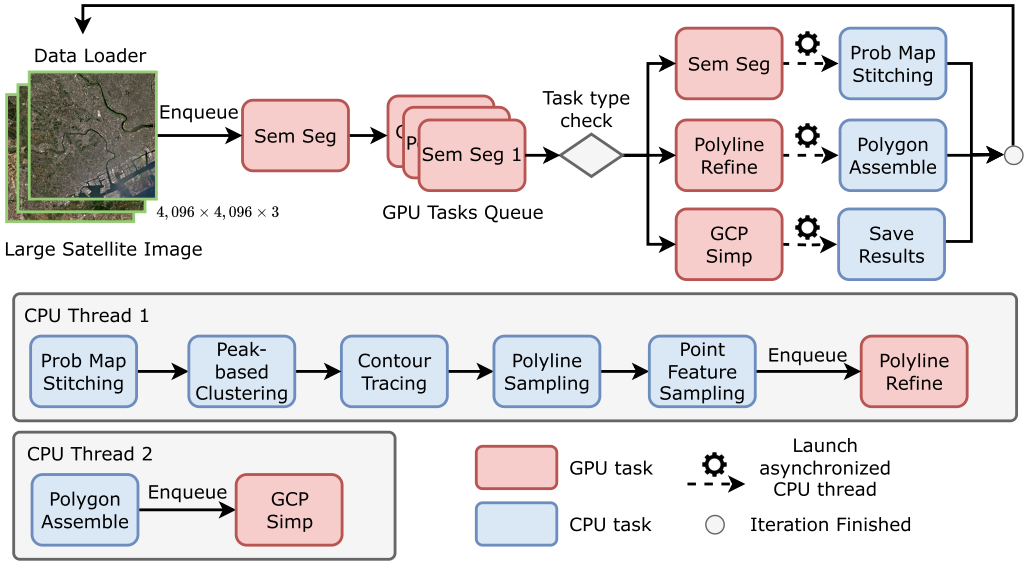


Fig. 5. Schematic of the proposed asynchronous inference pipeline. The workflow decouples GPU tasks—Sem Seg (semantic segmentation), polyline refinement, and GCP Simp (GCP simplification)—from CPU operations, including Prob map stitching, contour tracing, and polygon assembly. Satellite images @ Planet Labs PBC.

Oceania in both completeness and recall, indicating broader coverage and more effective detection.

Across Asia, our method consistently surpasses other datasets in completeness, IoU, precision, and recall, mainly because it includes China, which is absent in both GOB and MBF. In China specifically, our method outperforms CLSM on all metrics except MTA. This gap is likely due to the large number of tall buildings, where CLSM's off-nadir imagery focused on rooftops leads to greater alignment errors.

Because our evaluation is sensitive to geometric alignment, imagery accuracy also matters. PlanetScope has a geolocation error below 10 m RMSE, whereas the higher-resolution satellites used in GOB and MBF (e.g., WorldView-2/3) achieve CE90 below 4.1 m (approximately 2.7 m RMSE), partly explaining their stronger results in flat-building areas.

Overall, our method demonstrates competitive performance across diverse regions, particularly in Oceania and China, effectively filling coverage gaps left by existing global products.

2) Comparison With Learning-Based Baselines: The above comparison evaluates existing footprint products, but it does not directly assess whether our network design outperforms other polygon extraction algorithms at the model level. We therefore compare our method with FrameField [8] and HiSup [10] on the same global unified test set, as summarized in Table II and illustrated in Fig. 3.

As shown in Table II, our method achieves the best performance on all reported metrics except the N-ratio, where FrameField attains a value closer to unity. Overall, this indicates that our approach delivers more accurate polygon localization and better geometric regularization than the two VHR-oriented baselines under the medium-resolution setting.

Qualitative comparisons in Fig. 3 further highlight the differences. FrameField tends to produce polygons with overly dense vertices that are insufficiently simplified, and the predicted footprints often fail to align tightly with building boundaries. HiSup, by contrast, frequently generates shapes with unnatural geometry, including many sharp angles that do not resemble typical building footprints. Our method yields cleaner, more regularized polygons with better boundary fidelity despite the coarser input resolution.

D. Ablation Study

To evaluate the contribution of each component, we conduct an ablation study summarized in Table III.

We first test a naive baseline that directly thresholds and traces contours on the probability map. The first three rows show results under different binarization thresholds τ. Larger τ values yield thinner masks and fewer false merges, while

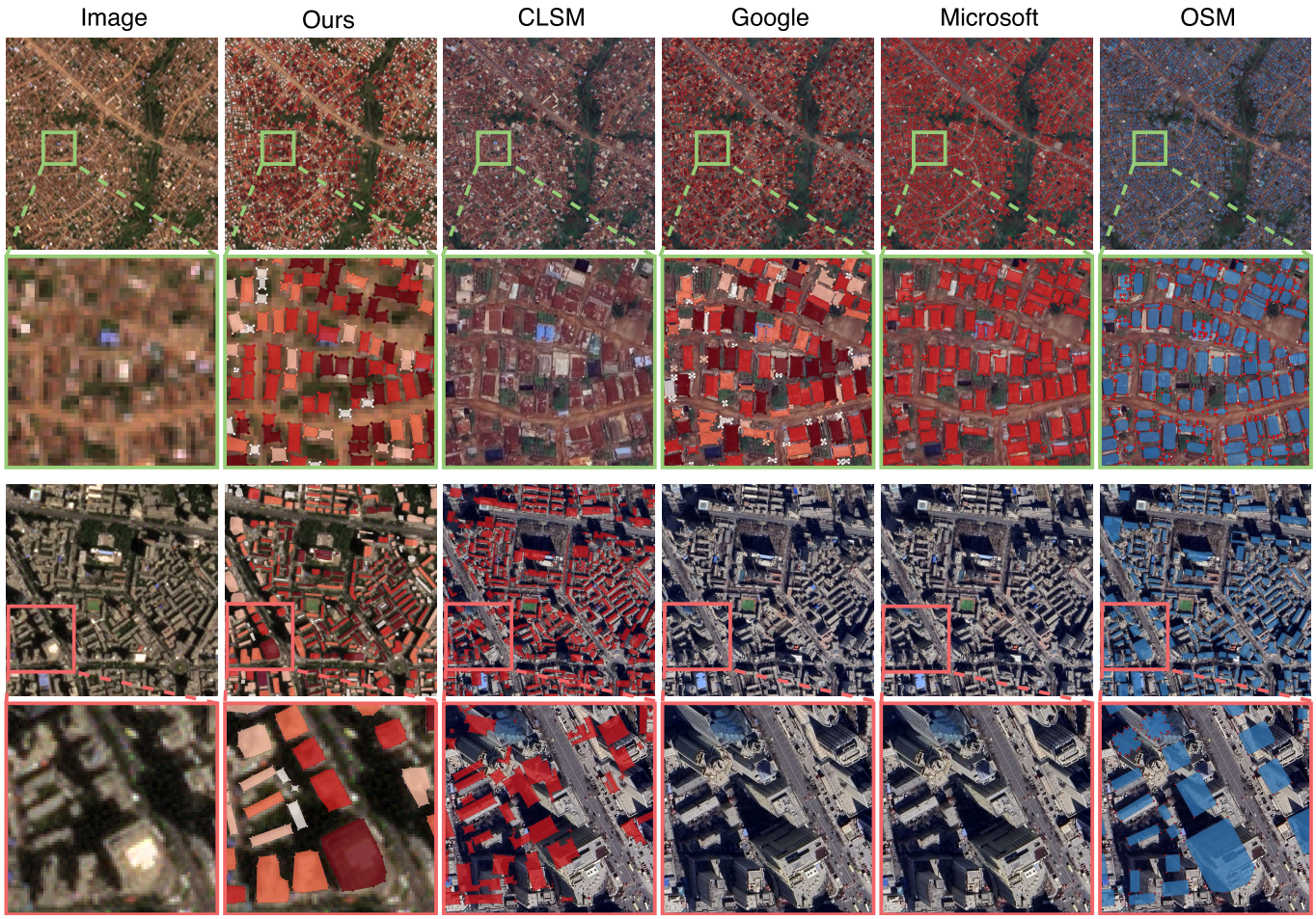


Fig. 6. Visual comparison of large-scale polygonal building footprint products, including GOB, MBF, and CLSM. Satellite imagery © Planet Labs PBC and Google Satellite; building annotations © OSM contributors. The upper and lower patches are sampled from Africa and China, respectively.

smaller ones retain more structures but cause oversegmentation. Among them, $\tau = 0.5$ achieves the best overall AP and AR, whereas $\tau = 0.4$ gives the highest IoU but with poor precision–recall balance.

Adding the PBC module substantially improves AP_{30}^{poly}, AR_{30}^{poly}, and N-ratio by enhancing instance separation, though with a slight drop in IoU and PoliS due to margin-induced shrinkage. Incorporating polyline refinement and GCP simplification [6] further enhances geometric regularity and alignment, leading to overall best performance and lower MTA values. The small decrease in AP_{50}^{poly} is mainly due to: 1) occasional over-splitting of large buildings by PBC and 2) reduced spatial precision for small buildings in medium-resolution imagery. The integration of the right-angle regularization loss $\mathcal{L}_{rht}$ further reduces MTA, indicating better angular alignment.

Visualization results under different ablation settings are shown in Fig. 4. The naive binarization method (third column) still merges small adjacent buildings even with a high threshold τ. After adding the PBC module (fourth column), instance separation improves notably. The final output with polyline regression and GCP simplification (fifth column) produces cleaner and more geometrically consistent polygons.

TABLE IV

IMPACT OF HYPERPARAMETERS

Ratio	ξ	AP_{50}^{poly}	AP_{30}^{poly}	N-ratio	MTA	PoliS	IoU
4×	0.05	3.1	15.4	0.59	34.4	3.8	38.4
8×	0.03	4.2	**18.3**	**0.64**	31.3	3.8	39.6
8×	0.05	4.5	**18.3**	0.60	31.5	3.5	40.5
8×	0.10	5.0	17.4	0.53	31.6	3.2	41.7
8×	0.20	**5.4**	15.8	0.43	31.4	3.0	42.8
8×	$+\infty$	5.0	12.6	0.34	**31.1**	**2.9**	**43.6**

E. Sensitivity Analyses

We evaluate the sensitivity of our method to two key hyperparameters in the PBC module: 1) the upsampling factor (via bilinear interpolation) and 2) the separation threshold ξ. Results are summarized in Table IV.

1) Effect of Upsampling Factor: Comparing the first and the third rows in the table, increasing the upsampling factor from 4× to 8× consistently improves all evaluation metrics. This indicates that a finer input resolution, even when generated using a naive upsampling algorithm, can enhance both instance segmentation accuracy and polygonal detail.

(a)

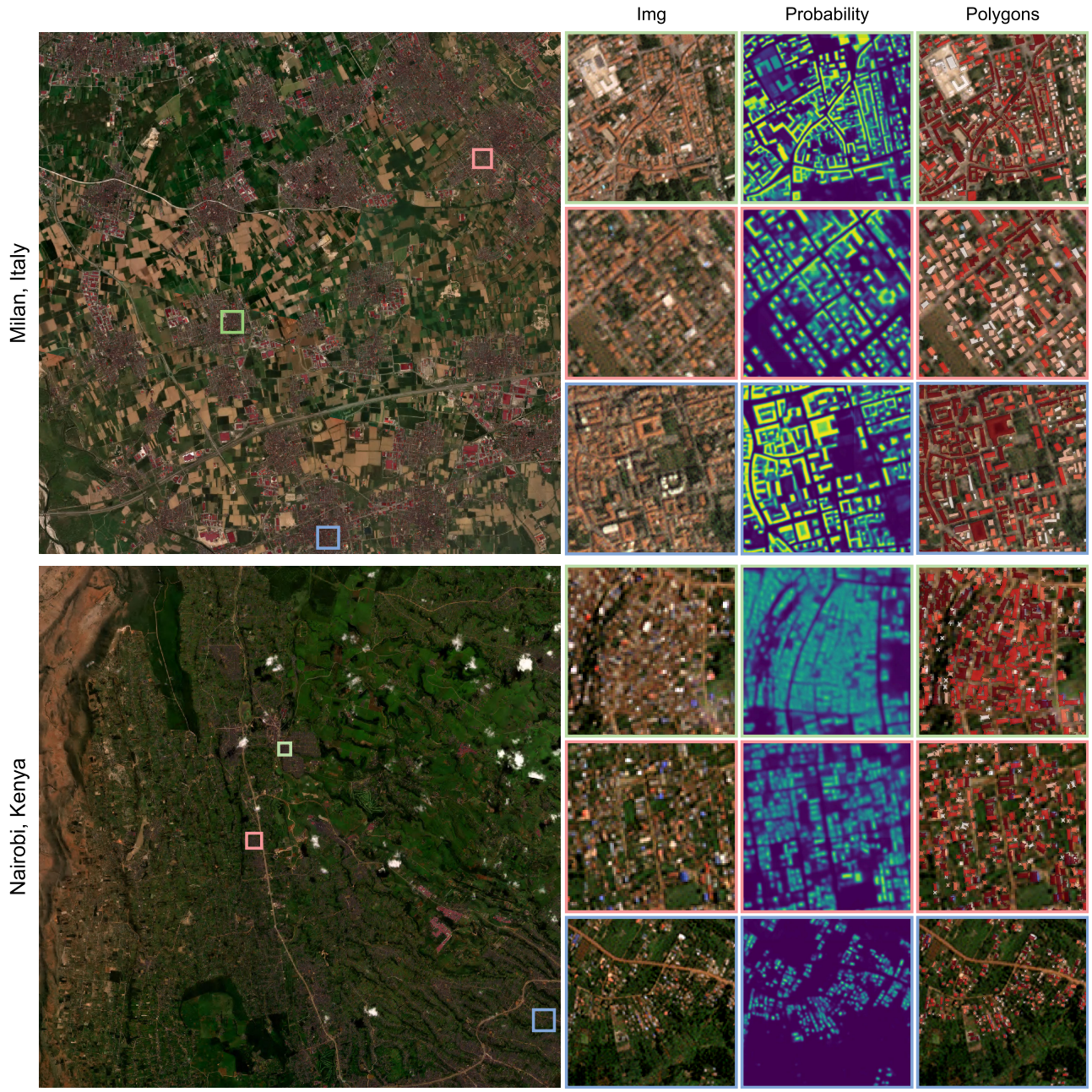
Fig. 7. Visualized comparison results with various large-scale polygonal building footprint products. Satellite images @ Planet Labs. (a) Visualization on two representative tiles sampled from Nanjing, China (Asia), and Vancouver, Canada (North America). (b) Visualization on two representative tiles sampled from Milan, Italy (Europe), and Nairobi, Kenya (Africa). (c) Visualization on two representative tiles sampled from Santiago, Chile (South America), and Melbourne, Australia (Oceania).

2) Effect of Threshold ξ in PBC: The separation threshold ξ controls the sensitivity of the PBC module to split adjacent buildings. As ξ increases, the model becomes less sensitive, making it less likely to divide large connected components into separate instances. This behavior leads to a lower N-ratio, indicating that fewer instances are produced. However, an excessively large ξ negatively affects Recall, IoU, and overall AP^{poly}. This degradation is primarily due to *under-segmentation*, where distinct adjacent buildings are erroneously merged into single polygons. While a smaller ξ might introduce slight boundary shrinkage due to aggressive splitting, a larger ξ sacrifices instance distinctiveness, leading to significant misses in densely built areas.

Overall, setting ξ in the range of 0.05–0.10 achieves a favorable balance, offering strong instance-level performance while maintaining accurate alignment with reference polygons.

F. Inference Efficiency

Table V presents the inference efficiency of our method in an asynchronous execution pipeline illustrated in Fig. 5.

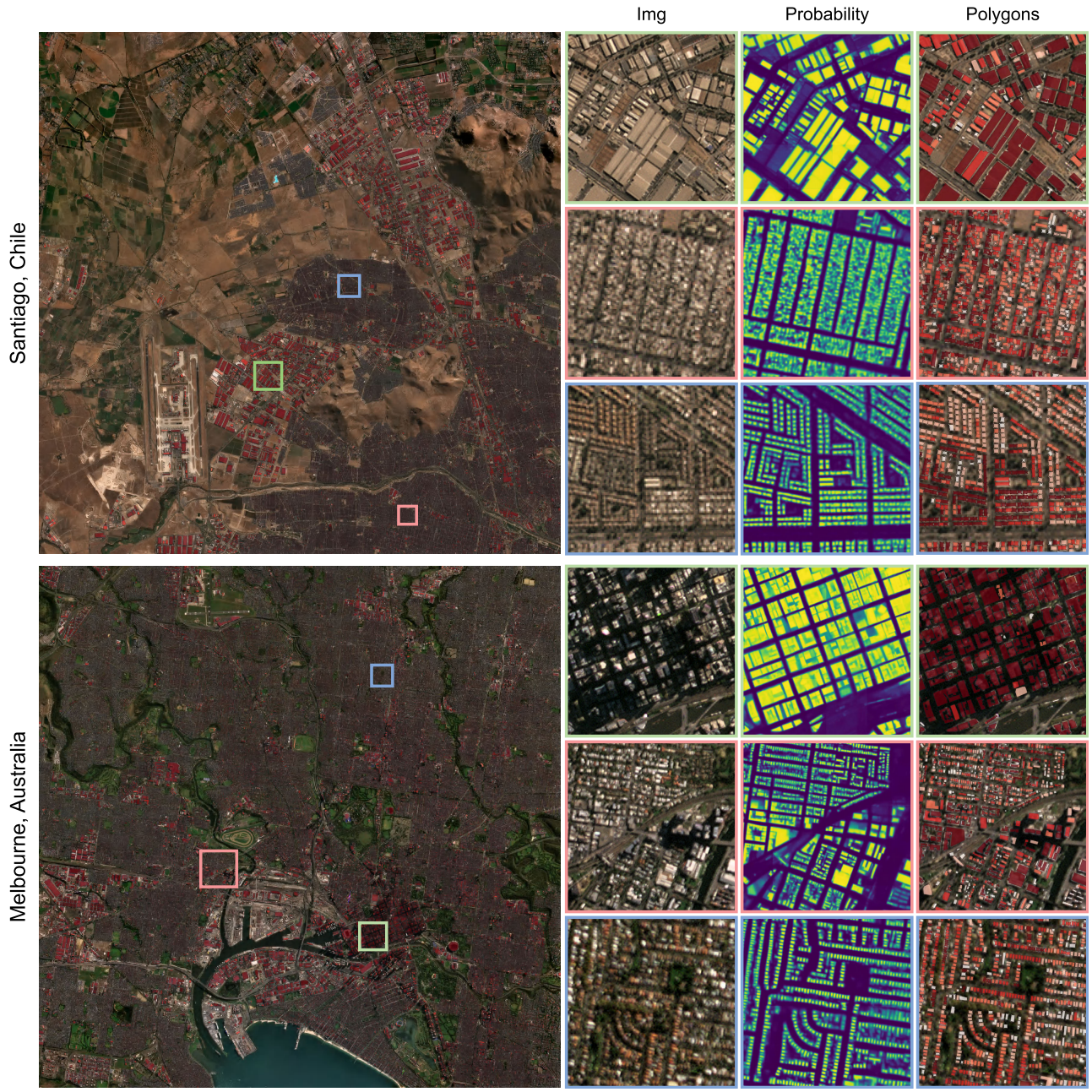


(b)

Fig. 7. *(Continued.)*Visualized comparison results with various large-scale polygonal building footprint products. Satellite images @ Planet Labs. (a) Visualization on two representative tiles sampled from Nanjing, China (Asia), and Vancouver, Canada (North America). (b) Visualization on two representative tiles sampled from Milan, Italy (Europe), and Nairobi, Kenya (Africa). (c) Visualization on two representative tiles sampled from Santiago, Chile (South America), and Melbourne, Australia (Oceania).

We evaluate two types of image tiles: "Urban" tiles sampled from regions defined in [36], and "Random" tiles uniformly selected from global locations. Each tile has a resolution of 4096 $\times$ 4096 pixels and is upsampled by a factor of eight during inference. For each category, we perform inference on 100 tiles and report the average runtime. We categorize the computation into GPU modules, which include semantic segmentation, polyline refinement, and GCP simplification, and CPU modules, which consist of probability map stitching, contour tracing, and polygon assembly. The total end-to-end inference time is also reported.

According to the results, semantic segmentation is the most computationally intensive among GPU modules, while refinement and simplification contribute less than 1% of the total time. In urban scenes, CPU-side processing becomes the primary bottleneck, taking nearly four times longer than GPU inference. Thanks to asynchronous scheduling and parallel CPU execution, the overall



(c)

Fig. 7. *(Continued.)*Visualized comparison results with various large-scale polygonal building footprint products. Satellite images @ Planet Labs. (a) Visualization on two representative tiles sampled from Nanjing, China (Asia), and Vancouver, Canada (North America). (b) Visualization on two representative tiles sampled from Milan, Italy (Europe), and Nairobi, Kenya (Africa). (c) Visualization on two representative tiles sampled from Santiago, Chile (South America), and Melbourne, Australia (Oceania).

pipeline achieves efficient inference, even in densely built environments.

G. Visual Comparison With Existing Products

We compare our predicted polygonal building footprints with three large-scale datasets: GOB, MBF, and CLSM. Two representative regions from Africa and China are selected for visual comparison, as shown in Fig. 6.

In terms of spatial coverage, GOB and MBF do not include China, while CLSM covers only parts of East Asia. Our method provides near-global coverage by using medium-resolution (4.8 m) imagery with a shorter revisit period, enabling more frequent and consistent updates.

In terms of footprint quality, GOB and MBF achieve superior geometric fidelity and angular precision owing to their use of high-resolution (0.5 m) imagery. Although our approach relies on coarser inputs, it still yields competitive performance. CLSM footprints are generally more regularized, often forming right-angled and simplified shapes. However, noticeable misalignments occur for tall buildings, likely caused by

TABLE V
AVERAGE LATENCY OF EACH COMPONENT IN THE PROPOSED INFERENCE PIPELINE WHEN PROCESSING A 4096 × 4096 IMAGE TILE (UPSAMPLED TO 32 768 × 32 768 DURING INFERENCE). "SEM. SEG." REFERS TO THE SEMANTIC SEGMENTATION MODEL, AND "POLY. RF." REFERS TO THE POLYLINE REFINEMENT MODULE

Type	Sem. Seg.	Poly. RF.	GCP	CPU	Overall
Urban	175.6s	1.6s	4.7s	643.7s	185.2s
Random	152.7s	0.2s	0.2s	50.9s	153.9s

off-nadir viewing angles that displace rooftops. Our method uses near-nadir medium-resolution imagery, which reduces parallax effects and produces more accurately aligned footprints in dense urban areas.

H. Visualization Results on Large Satellite Images

The proposed polygonal building mapping pipeline demonstrates strong capability in processing large-scale satellite scenes. Fig. 7 presents visualizations on 4096 × 4096 image tiles sampled across six continents. We analyze performance across different types of geographic regions.

1) Structured Urban Regions (e.g., Nanjing, Vancouver, Melbourne): In areas with well-aligned and uniformly distributed buildings, our method consistently produces accurate polygons. Despite the moderate resolution, the results exhibit strong alignment and completeness, demonstrating robustness in structured urban environments.

2) Dense and Irregular Urban Regions (e.g., Milan): In areas where buildings are densely clustered and often form large, irregularly shaped complexes, the model may under-segment these connected structures. While the probability maps still capture detailed local boundaries, the polyline refinement module sometimes fails to produce accurate polygon delineations. A likely reason is the limited availability of well-matched predicted and ground-truth polygon pairs for such complex geometries during training, which reduces the effectiveness of supervision in these cases.

3) Small and Scattered Buildings (e.g., Nairobi): In regions such as Nairobi where buildings are extremely small, sometimes as small as 2 × 2 pixels, the model can generate reasonably detailed probability maps. However, due to resolution limitations, it may struggle to separate adjacent buildings when they are densely clustered.

4) Tiling Artifacts at Patch Boundaries: Due to the sliding-window inference strategy, tiling artifacts may occur at the borders of adjacent patches. These artifacts, caused by inconsistencies in predicted probabilities across tile boundaries, can lead to incorrect instance splitting in the PBC module. For example, in the Santiago scene, border fluctuations in the probability map result in some large buildings being erroneously divided.

Overall, our method generalizes well across diverse global landscapes. While it faces challenges in highly dense or extremely fine-scale regions, it remains efficient and scalable. Combined with the frequent revisit cycles of medium-resolution satellites, it presents a promising approach to maintaining up-to-date global building footprints.

V. CONCLUSION

This article presents a scalable framework for extracting building polygons from medium-resolution satellite imagery at a global scale. We build a large-scale dataset by aligning PlanetScope basemaps with building annotations from multiple public sources, ensuring broad geographic diversity. To improve instance delineation and scalability in semantic segmentation-based extraction, we propose a PBC method for accurate separation of adjacent buildings and a PFS strategy for efficient polygon generation on large tiles. Experiments show that our method achieves state-of-the-art accuracy in China and provides broader coverage and more frequent updates elsewhere, while maintaining competitive performance against existing global products. Beyond accuracy, our approach enhances the completeness of global building polygons, particularly in regions underrepresented in high-resolution datasets, making it a valuable complement to current solutions and a practical tool for downstream applications.

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Fahong Zhang received the B.E. degree in software engineering from Northwestern Polytechnical University, Xi'an, China, in 2017, and the M.S. degree in computer science with the Center for OPTical IMagery Analysis and Learning, Northwestern Polytechnical University, Xi'an. He is currently pursuing the Ph.D. degree with the Department of Aerospace and Geodesy, Data Science in Earth Observation, Technical University of Munich, Munich, Germany.

His research interests include computer vision and satellite image processing.

Xiao Xiang Zhu (Fellow, IEEE) is the Chair Professor for Data Science in Earth Observation at the Technical University of Munich (TUM), Munich, Germany, where she also serves as the Director of Munich Data Science Institute (MDSI). She is currently a Visiting AI Professor at European Space Agency's ϕ-Laboratory, Frascati, Italy. Since 2022, she has led the National Center of Excellence ML4Earth (ml4earth.de), funded by German Space Agency. Her research interests include Earth observation and remote sensing, signal processing, machine learning, and data science, with applications to global urbanization, the UN sustainable development goals, and climate change. For more information, see https://www.asg.ed.tum.de/sipeo